CSC-103 (Discrete Structures) Syllabus

General Information

Course Number	CSC-103
Credit Hours	3+0=3 (Theory credit hours = 3; Lab. credit hours = 0; Total credit hours = 3)
Prerequisite	None
Course Coordinator	Mr. Rabnawaz Mallah

COURSE OBJECTIVE:

The main focus of this course is to understand a particular set of mathematical facts and how to apply them; more importantly, such a course should teach students how to think logically and mathematically. This course introduces a formal system (propositional and predicate logic) on which mathematical reasoning is based. It develops an understanding of how to read and construct valid mathematical arguments (proofs) and mathematical statements (Theorems). It also develops the ability to see a problem from a mathematical perspective. The discrete structure course introduces various problem-solving strategies, especially thinking algorithmically (both iterative and recursive). This course covers the discrete data structures such as sets, relations, discrete functions, graphs, and trees

	Course Learning Outcomes (CLO)
1	Understand the key concepts of discrete structures such as sets, permutations, relations, graphs and trees.
2	Apply these concepts to solve computational problems.
3	Apply mathematical reasoning in order to read, comprehend, and construct mathematical arguments.
4	Develop mathematical skills by practicing problem solving, modeling, logical reasoning and writing precise proofs.

COURSE WHITAGES/ASSESSMENT

		SESSIONAL MARKS 30%				
S.N0	**ASSESSMENTS ACTIVITIES**	**QUIZ-1 10 Marks** **QUIZ-1 10 Marks**	**ASSIGNMNET-1** **ASSIGNMNET-2**	**MID 30 Marks**	**FINAL 30 Marks**	**TOTAL**
1	**Discrete Structure (CSC-103)**	**20%**	**10%**	**30%**	**40%**	**100%**

Recommended Books for Discrete Mathematics

1	Kenneth H. Rosen, “Discrete Mathematics and its Applications”, Seventh
2	Discrete Structure by Richard Johnson Baugh “ 7th Edition”
3	Discrete Structure. “ Virtual University”
4	Discrete Structure. “ NESO PPTS”

COURSE OUTLINES

Weeks	LEC#	SUB TOPICS	CLOS	PLOS
1	Lec # 1	Introduction of Discrete Structure	1	1
	Lec # 2	Propositional Logic	1	1
	Lec # 3	Application of Propositional Logic	2	2
2	Lec # 4	Propositional Logic, Propositional variables, Compound Propositions	2	2
	Lec # 5	Negation, Conjunction and Disjunctions with Truth Tables	3	2
	Lec # 6	Exclusive OR, Implication and Bi-Conditional with Truth Tables	3	2
3	Lec # 7	QUIZ-1		
	Lec # 8	Propositional Equivalences inferences and proofs	3	5
	Lec # 9	Tautology, Contradiction, Contingency, Satisability	4	2
4	Lec # 10	Assignment-1 (Rules of Inference, Introduction to Proofs)		
	Lec # 11	Predicate and Quantifiers	2	2
	Lec # 12	Types of Quantifiers	2	2
5	Lec # 13	Nested Quantifiers Case-1 & 2	2	2
	Lec # 14	Nested Quantifiers Case-3 & 4	2	2
	Lec # 15	QUIZ-2		
6	Lec # 16	Translation in English and vice versa	1	1
	Lec # 17	Introduction of Set, Types of set, Method of Sets	1	1
	Lec # 18	Set Operations with Builder Notations & Cardinality of Set	1	1
7	Lec # 19	Properties od Sets with Venn Diagram	2	1
	Lec # 20	Function and Types of Function	2	2
	Lec # 21	Recursive Functions, Sequences and Summations	2	2
8	Lec # 22	The Integers and Division, Primes and Greatest Common Divisors	1	2
	Lec # 23	Mathematical Induction	2	2
	Lec # 24	Matrices	3	2
9	Lec # 25	The Basics of Counting,	2	2
	Lec # 26	The Pigeonhole Principle,	2	2
	Lec # 27	Permutations and Combinations,	3	2

10	**Lec # 34**	Binomial Coefficients (Expansion)	3	2
	Lec # 35	Binomial Coefficients (Series)	2	2
	Lec # 36	Recurrence Relations,	3	2
11	**Lec # 37**	**Assignment-2**		
	Lec # 38	Solving Linear Recurrence Relations,	3	3
	Lec # 39	Inclusion-Exclusion, Applications of Inclusion-Exclusion	3	3
12	**Lec # 40**	Relations and Their Properties,	3	3
	Lec # 41	Representing Relations,	3	3
	Lec # 42	Types of Relations	2	2
13	**Lec # 43**	Equivalence Relations	3	4
	Lec # 44	Partial Ordering Relations	3	4
	Lec # 45	Introduction of Graphs	1	2
14	**Lec # 46**	Types of the Graphs	2	2
	Lec # 47	Graph Terminology	1	1
	Lec # 48	Degree of the Graph	2	2
15	**Lec # 49**	Path and Circuits	3	2
	Lec # 50	Matrix Representation of Graphs	3	3
	Lec # 51	Isomorphic Graphs	3	3
16	**Lec # 52**	Trees and Properties of Trees	3	3
	Lec # 53	Representation of Algebraic expressions by Binary Trees	3	4
	Lec # 54	Spanning Trees	2	4

Approved by

Subject Teacher Name	Rabnawaz Mallah
HOD	Abdul Haseeb Shaikh
Chairman	Dr. Syed Rehan Ali Shah
Dated	01-01-2025